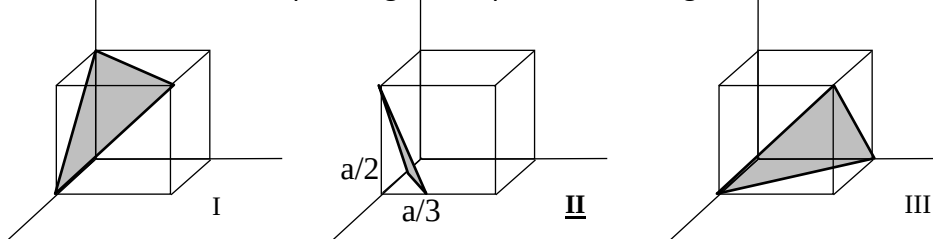


Problems on crystalline structures

1. Find the Miller indices corresponding to the planes in the figures.



For an FCC structure, with lattice parameter a , calculate the linear density along direction $[110]$, the planar density on the plane drawn on figure I and the volumetric density.

(Solution: I $(\bar{1}\bar{1}\bar{1})$; II $(\bar{2}31)$; III $(\bar{1}\bar{1}1)$; $\rho_l = \sqrt{2}/a$, $\rho_s = 4/a^2 \sqrt{3}$, $\rho_v = 4/a^3$)

2. Determine the atomic mass of a metal element with FCC structure, density of 1.74 g/cm^3 and lattice constant of $4.527 \text{ \AA}$. Find the element with these characteristics.

(Solution: $M=24.3 \text{ g/mol}$; Mg)

3. Draw the crystalline planes in a cubic lattice that present the following Miller indices:

- | | | | | |
|------------------|------------------|------------------------|------------------------|------------------------|
| a) $(10\bar{1})$ | b) $(1\bar{2}1)$ | c) $(\bar{2}13)$ | d) $(1\bar{3}3)$ | e) $(1\bar{2}\bar{2})$ |
| f) $(\bar{3}12)$ | g) $(12\bar{3})$ | h) $(\bar{1}\bar{4}3)$ | i) $(\bar{3}1\bar{3})$ | j) $(3\bar{1}3)$ |

4. For a BCC structure, determine the surface (planar) density for the planes (100) , (110) and (111) ; the linear density in the direction $[100]$, and the volume density.

(Solution: $\rho_{(100)} = 1/a^2$, $\rho_{(110)} = \sqrt{2}/a^2$, $\rho_{(111)} = 1/a^2 \sqrt{3}$, $\rho_{[100]} = 1/a$, $\rho_v = 2/a^3$)

5. Given that aluminum has an FCC structure, the atomic radius $1.43 \text{ \AA}$ and the atomic weight 26.98 g/mol , calculate:

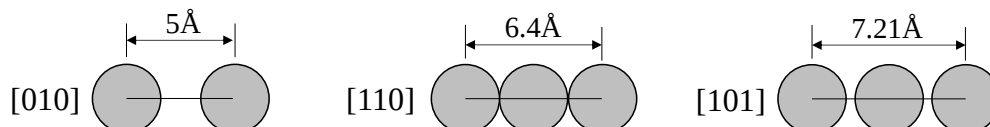
- Planar density, in at/cm^2 , for the (110) plane. **(Solution: $\rho_{(110)} = 8.7 \times 10^{14} \text{ at/cm}^2$)**
- Linear density, in at/cm , for the $[110]$ direction. **(Solution: $\rho_{[110]} = 3.5 \times 10^7 \text{ at/cm}$)**
- Weight of a bar of 20 mm in diameter and 1 m long. **(Solution: $m = 855 \text{ g}$)**

Data: $1 \text{ m} = 10^{10} \text{ \AA}$; $N_A = 6.023 \times 10^{23} \text{ mol}^{-1}$.

6. Calculate the linear density in the $[111]$ direction and the planar density in the (111) plane for:

- Iron BCC **(Solution: $\rho_{[111]} = 2/\sqrt{3}a$, $\rho_{(111)} = 1/a^2 \sqrt{3}$)**
- Nickel FCC **(Solution: $\rho_{[111]} = 1/a\sqrt{3}$, $\rho_{(111)} = 4/a^2 \sqrt{3}$)**

7. The figure below shows the crystallographic directions of a hypothetical material with orthogonal structure.



- Draw the unit cell. To which crystalline lattice does it belong?
- Calculate the atomic weight given that the density is 5.97 g/cm^3 .
- Calculate the planar density in atoms/ mm^2 in the planes (100) and (110), and compare them with each other.

(Solution: a) face centered orthorhombic lattice; b) $M=107.87 \text{ g/mol}$; c) $\rho_{(100)}=6.66 \times 10^{12} \text{ at/mm}^2$; $\rho_{(110)}=5.2 \times 10^{12} \text{ at/mm}^2$)

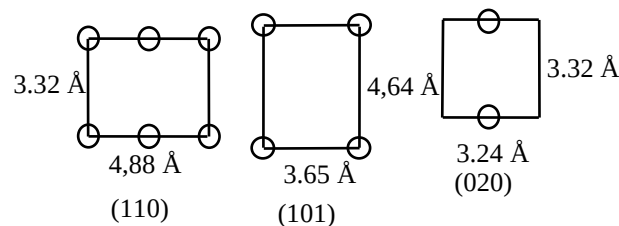
- Calculate the fraction of area occupied by atoms for the (111), (200), (220), (222), (400) and (420) planes, in the FCC structure.

(Solución: (111): $\pi/2\sqrt{3}$; (200): $\pi/4$; (220): $\pi\sqrt{2}/8$; (222): 0; (400): 0; (420): $\pi/4\sqrt{5}$)

- A pure metal undergoes a polymorphic change from BCC to FCC when it reaches 910°C . Calculate the volume change associated with the change in structure given that the interplanar spacing d_{321} for the BCC structure is 0.07565 nm , and the planar density in the FCC lattice in the plane (002) is $15.18 \cdot 10^{18} \text{ at/m}^2$.

(Solution: 5.7 %)

- Three distinct crystallographic planes in the unit cell of a hypothetical metal are shown below. The circles represent atoms.



- To which crystalline structure does this unit cell belongs to?
- Given that the density of this metal is 5.9 g/cm^3 , determine the atomic mass. Which specific metal is it referred to?
- Identify the directions and planes of maximum packing.

(Solution: a) Base centered orthorhombic lattice; b) $M= 69.76 \text{ g/mol}$, Gallium; c) $[\bar{1}10], [\bar{1}\bar{1}0], [\bar{1}10], [\bar{1}\bar{1}0], (001)$)

- Aluminum has an FCC structure with atomic weight of 26.98 g/mol . Given that its linear density in the direction $[111]$ is $1.43 \cdot 10^7 \text{ at/cm}$, determine:

- The lattice parameter and the atomic radius for aluminum.
- Volume density in (g/cm^3) and planar density in the plane (101) in (at/cm^2).

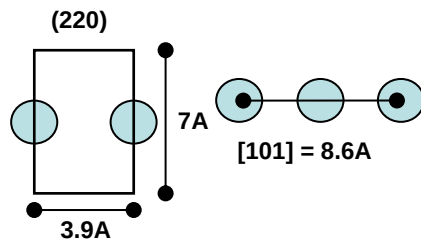
(Solution: a) $a= 4.04 \text{ Å}$, $r=1.43 \text{ Å}$; b) $\rho = 2.71 \text{ g/cm}^3$; $\rho_{(101)}=8.66 \cdot 10^{14} \text{ at/cm}^2$)

12. Draw direction vectors in unit cells for the following cubic directions

- a) $[1\bar{1}2]$ b) $[\bar{3}31]$ c) $[2\bar{1}2]$ d) $[\bar{1}01]$ e) $[321]$ f) $[1\bar{2}2]$ g) $[1\bar{2}3]$ h) $[0\bar{2}1]$ i) $[2\bar{3}3]$ j) $[12\bar{1}]$ l) $[10\bar{3}]$
k) $[\bar{2}23]$.

13. In the figure, the dimensions of the (220) plane and the $[101]$ direction of a hypothetical metal with a radius of $1.98 \text{ \AA}$ and atomic weight of 180.95 g/mol are shown.

- Draw the unit cell, deduce which crystal system it belongs to, and identify the name of this crystal structure.
- Determine the packing factor.
- Calculate the density on the (110) and (100) planes in atoms/cm^2 .
- Calculate the interplanar distances $d(111)$ and $d(110)$ in $\text{\AA}$.
- Calculate the weight of a single crystal with a volume of 1 cm^3 .



(Solution: a) Face centred Orthorhombic b) 0,62 c) $\rho_{(110)} = 3.66 \cdot 10^{14} \text{ atoms/cm}^2$; $\rho_{(100)} = 4.76 \cdot 10^{14} \text{ atoms/cm}^2$ d) $d_{(111)} = 3.37 \text{ \AA}$; $d_{(110)} = 3.84 \text{ \AA}$ e) 5,72gr)

14. Fe has a BCC or an FCC structure, depending on the temperature. The X-ray diffraction (XRD) technique allows determining the spacing or distances between crystalline planes, from which the lattice parameters are deduced. For the BCC structure, the lattice parameter is $a = 0.2864 \text{ \AA}$, while for the FCC $a = 0.3592 \text{ \AA}$.

- What distance can be expected between the planes (020) for both structures?
- Calculate the distance between the most compact planes of both structures.

(Solution: a) $d_{020} \text{ (BCC)} = 0.143 \text{ \AA}$; $d_{020} \text{ (FCC)} = 0.179 \text{ \AA}$; c) BCC (110) = $0.2025 \text{ \AA}$; FCC (111) = $0.207 \text{ \AA}$)